

Expt No:

03

TRAFFIC LIGHT CONTROL SYSTEM

Date:

/ /2025

1. AIM:

To implement a one way and two-way traffic light control system.

2. ONE WAY DIRECTION:

2.1. PROBLEM STATEMENT:

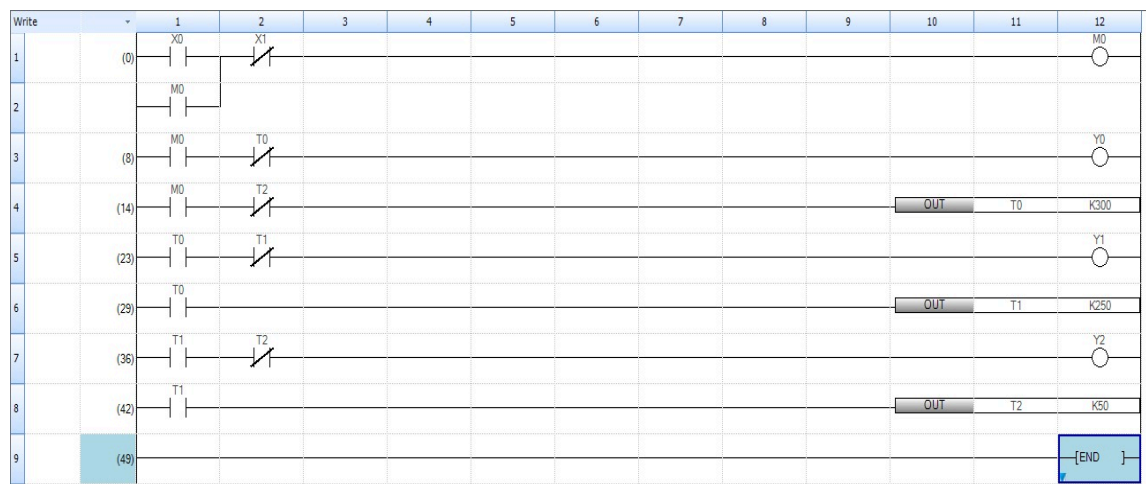
Red light glows for 30s, yellow light for 5 s, green light for 25s and this process continues. There is a start button and a stop button. Draw the ladder logic to implement the above process.

2.2. FLOW DIAGRAM:



Figure 2.1. One way traffic light control system.

2.3. LADDER DIAGRAM:



3. TWO-WAY DIRECTION:

3.1. PROBLEM STATEMENT:

When Red 1 is ON for 30s, E-W traffic flows. At this instant, Yellow 2 becomes ON for 5s after which Green 2 becomes ON for 25s. After 30s, when Red 2 is ON for 30s, N-S traffic flows. At this instant, Yellow 1 becomes ON for 5s after which Green 1 becomes ON for 25s and this process continues. There is a start button and a stop button. Draw the ladder logic to implement the above process.

3.2. FLOW DIAGRAM:

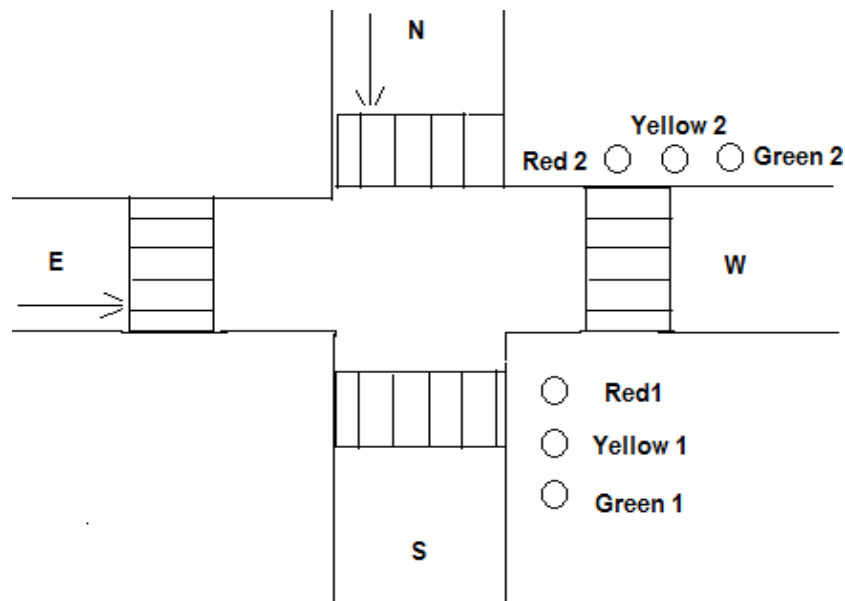
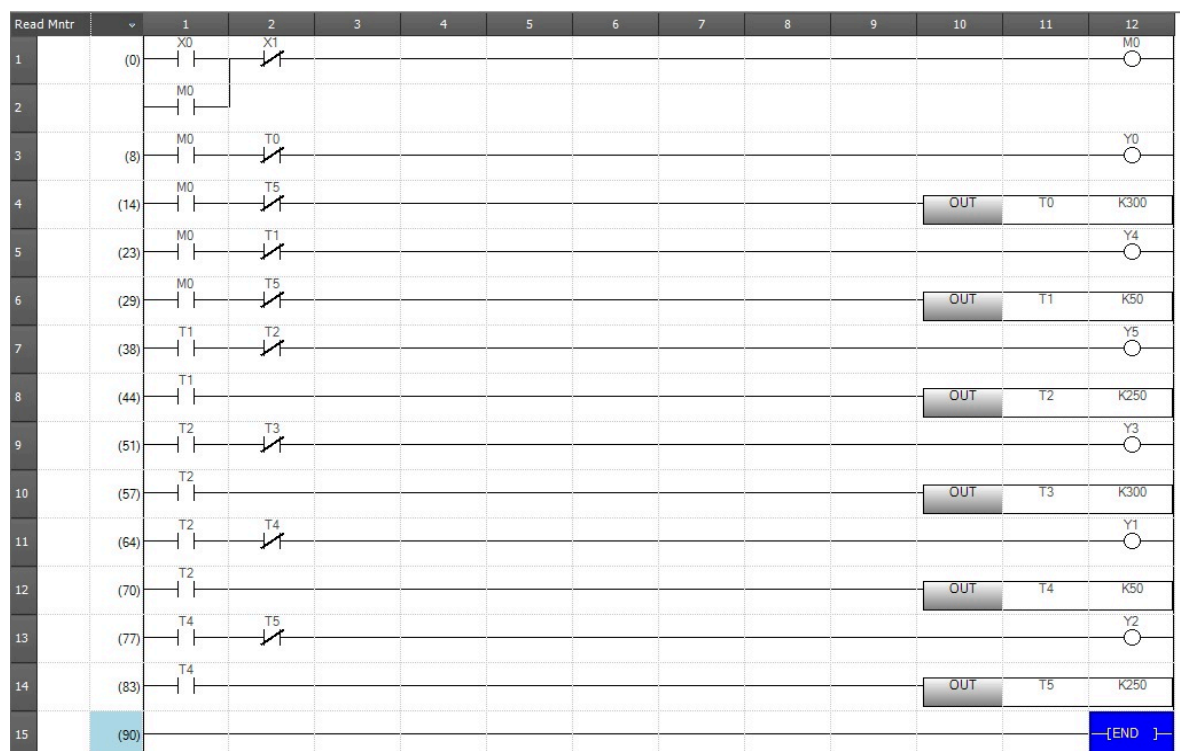


Figure 3.1. Two way traffic light control system.

3.3. LADDER DIAGRAM:



4. **RESULT:**

Thus, the one way and two-way traffic light control was executed and verified using PLC.